

Name _____

Period _____ Date _____

Practice Test – Advanced Kinematics (Version 3)

On any problems, you must show all work, include units, round properly, and box your final answer.

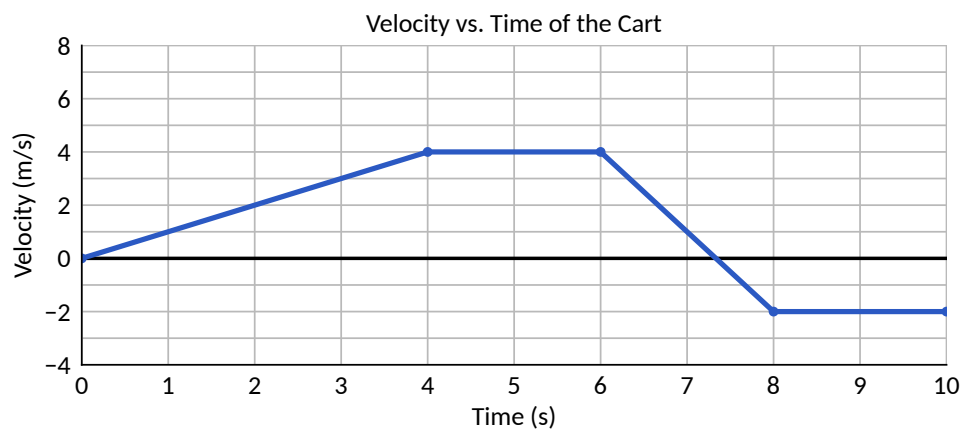
Do the algebra first, before plugging in values.

Rewrite the following statement in the space below: On my honor, before God, I have neither given, nor received, any aid on this test.

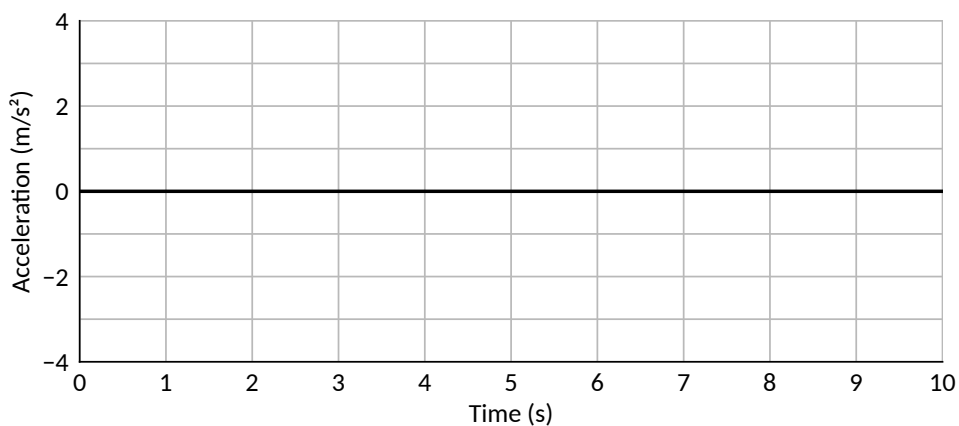
For all questions:

- Use $g = 9.80 \text{ m/s}^2$ and neglect air resistance.
- Round numerical answers to three significant figures.
- 50 minutes. Four questions, 25 points each (100 points total).

1. (25 points) The velocity-time graph below shows the motion of a cart moving along a straight track from $t = 0$ s to $t = 10$ s. On the blank grid, draw the corresponding acceleration-time graph. Show the slope calculation for each segment of the velocity-time graph.



Draw your acceleration-time graph here:



Slope calculations:

2. (25 points) *A car traveling at 30.0 m/s on a straight, level road brakes with a constant acceleration of magnitude 7.50 m/s^2 until it comes to a stop.*

a. **(15 points)** How far does the car travel while braking?

b. **(5 points)** How long does it take the car to stop?

c. **(5 points)** If the car's initial speed were doubled and it braked with the same acceleration, by what factor would its braking distance change? Justify your answer using your expression from part (a).

3. (25 points) A football is kicked from ground level on a level field with an initial speed $v_0 = 28.0 \text{ m/s}$ at an angle $\theta = 30.0^\circ$ above the horizontal. It lands at the same height from which it was kicked.

a. **(15 points)** Derive an expression for the ball's range R in terms of v_0 , θ , and g . Then calculate the range.

b. **(5 points)** What is the ball's hang time?

c. **(5 points)** What is the ball's maximum height above the field?

4. (25 points) A small object is released from rest at the top of a skyscraper of height H above the ground. Air resistance is negligible. Choose upward as the positive direction.

a. **(15 points)** Derive an expression for the object's velocity just before it hits the ground, including its sign, in terms of H and g .

b. **(10 points)** Write an expression for the object's velocity $v(t)$ as a function of the time t since it was released, and state the time interval over which this expression is valid, in terms of H and g .

Answer Key – Practice Test, Version 3 – Advanced Kinematics

Worked step by step, the same way as the class notes: sketch, list the variables, pick the equation, algebra first, then numbers with units. $g = 9.80 \text{ m/s}^2$; answers to three significant figures.

AP Physics 1 equation sheet, kinematics (Mechanics and Fluids, first three):

#1 $v_x = v_{x0} + a_x t$ · **#2** $x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$ · **#3** $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$

Pick the one that leaves out the variable you don't know and don't need. For vertical motion use the same three with y . $x - x_0$ is the displacement Δx .

1. Acceleration-time graph (25 points)

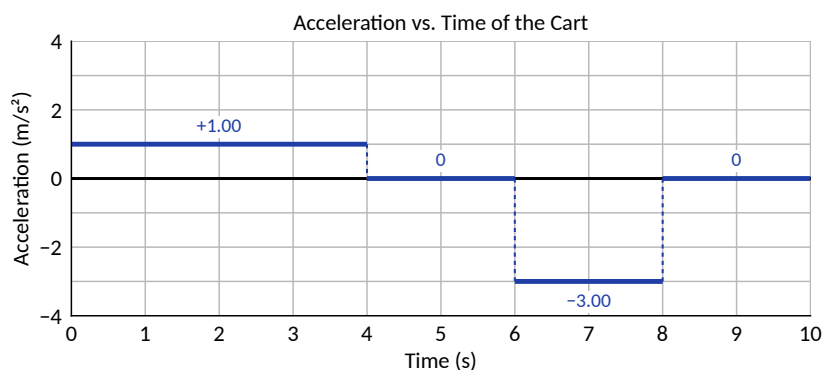
Step 1 Acceleration is the slope of the v-t graph.

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{t_f - t_i} \text{ (read } v_i \text{ and } v_f \text{ at the start and end of each straight piece)}$$

Step 2 Find the slope of each segment.

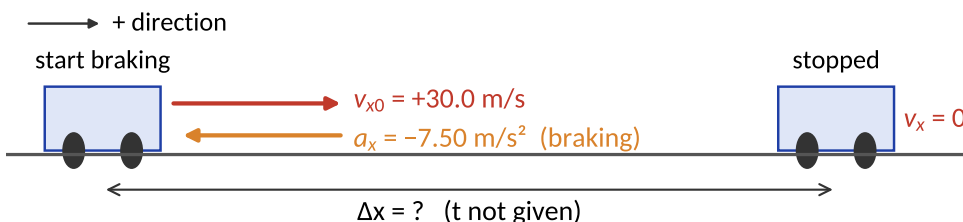
segment	$\Delta v = v_f - v_i$	$\Delta t = t_f - t_i$	$a = \Delta v / \Delta t$
0 s to 4 s	4 m/s - 0 m/s = 4 m/s	4 s - 0 s = 4 s	+1.00 m/s ²
4 s to 6 s	4 m/s - 4 m/s = 0 m/s	6 s - 4 s = 2 s	0 m/s ²
6 s to 8 s	(-2 m/s) - 4 m/s = -6 m/s	8 s - 6 s = 2 s	-3.00 m/s ²
8 s to 10 s	(-2 m/s) - (-2 m/s) = 0 m/s	10 s - 8 s = 2 s	0 m/s ²

Step 3 Draw each slope as a flat line over its time interval. Label both axes with units.



Watch out: the cart turns around where v crosses 0, but the slope does not change there, so a does not change either. A straight slanted v-t line always becomes a flat a-t line.

2. Braking car (25 points)



(a) Braking distance (15 points)

Step 1 Pick a positive direction.

Direction of motion is +. Braking points the other way, so a is negative.

Step 2 List the five variables (the way they are written on the AP equation sheet).

v_{x0} (initial)	v_x (final)	a_x	t	$x - x_0 = \Delta x$
+30.0 m/s	0 m/s (stops)	-7.50 m/s ²	not given, not asked	? (wanted)

Step 3 Pick the AP equation sheet equation that leaves out the variable you don't have and don't need.

AP sheet	equation	leaves out
#1	$v_x = v_{x0} + a_x t$	$x - x_0$ (Δx)
#2	$x = x_0 + v_{x0}t + \frac{1}{2}a_x t^2$	v_x (final velocity)
#3	$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	t (time) ← use this one

Why #3: t is not given and part (a) does not ask for it. #3 is the only kinematic equation without t, and every other variable in it is known except Δx .

Step 4 Solve with letters first.

$$v_x^2 = v_{x0}^2 + 2a_x(x - x_0) \rightarrow 0 = v_{x0}^2 + 2a_x\Delta x \rightarrow \Delta x = -\frac{v_{x0}^2}{2a_x}$$

Step 5 Plug in numbers with units.

$$\Delta x = -\frac{(+30.0 \text{ m/s})^2}{2(-7.50 \text{ m/s}^2)} = -\frac{(900 \text{ m}^2/\text{s}^2)}{(-15.0 \text{ m/s}^2)} = \boxed{60.0 \text{ m}} \text{ (the negatives cancel, so the distance is positive)}$$

(b) Stopping time (5 points)

Step 1 Pick the equation.

AP sheet	equation	leaves out
#1	$v_x = v_{x0} + a_x t$	$x - x_0$ (Δx) ← use this one
#2	$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$	v_x (final velocity)
#3	$v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$	t (time)

Why #1: now t is wanted, and Δx is not needed. #1 leaves out Δx and has only t unknown.

Step 2 Solve with letters, then plug in.

$$v_x = v_{x0} + a_x t \rightarrow 0 = v_{x0} + a_x t \rightarrow t = -\frac{v_{x0}}{a_x} = -\frac{(+30.0 \text{ m/s})}{(-7.50 \text{ m/s}^2)} = \boxed{4.00 \text{ s}}$$

(c) Double the initial speed (5 points)

Step 1 Look at the expression from part (a).

$$\Delta x = -\frac{v_{x0}^2}{2a_x} \rightarrow v_{x0} \text{ is squared, and } a_x \text{ stays the same.}$$

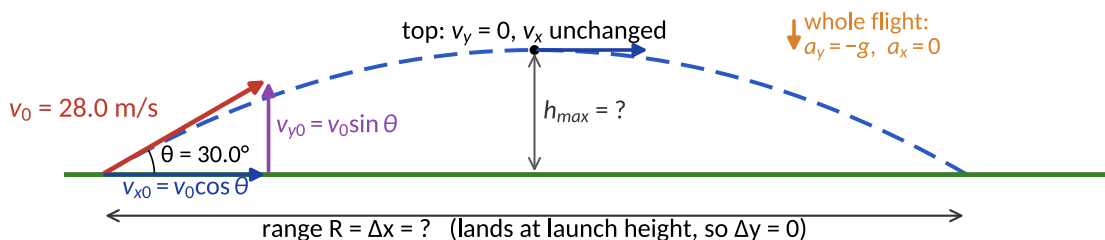
Step 2 Replace v_{x0} with $2v_{x0}$.

$$\Delta x' = -\frac{(2v_{x0})^2}{2a_x} = -\frac{4v_{x0}^2}{2a_x} = 4\Delta x \rightarrow \text{the braking distance is multiplied by } \boxed{4.00}$$

Check: $4(60.0 \text{ m}) = 240 \text{ m}$.

Watch out: doubling the speed does not double the distance. Because v_{x0} is squared, the distance goes up by $2^2 = 4$.

3. Football range (25 points)



Big idea: split the kick into two motions. Sideways moves at a constant speed ($a_x = 0$). Up and down is free fall ($a_y = -g$). **Time** is the only thing both share.

(a) Derive the range, then calculate it (15 points)

Step 1 Split v_0 into horizontal and vertical parts. Keep them as letters; numbers go in at the end.

$$v_{x0} = v_0 \cos \theta \text{ (cos goes with the side next to the angle: the ground)}$$

$$v_{y0} = v_0 \sin \theta \text{ (sin goes with the side across from the angle: straight up)}$$

Step 2 Make the T-chart.

horizontal	vertical
$a_x = 0$	$a_y = -g = -9.80 \text{ m/s}^2$
$v_{x0} = v_0 \cos \theta$	$v_{y0} = v_0 \sin \theta$
$\Delta x = R = ?$	$\Delta y = 0$ (lands at launch height)
$t = ?$ (same t)	$t = ?$ (same t)

Step 3 Get the time from the vertical column (you know Δy there).

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity) ← use this one
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time)

Why #2: in the vertical column you know $\Delta y = 0$, v_{y0} and a_y , and you want t . The landing velocity v_y is not given, so use #2, the one that leaves out v_y .

$$y - y_0 = v_{y0} t + \frac{1}{2} a_y t^2 \rightarrow 0 = (v_0 \sin \theta) t - \frac{1}{2} g t^2 \rightarrow 0 = t(v_0 \sin \theta - \frac{1}{2} g t)$$

$t = 0$ is the kick, so the landing is at $t = \frac{2v_0 \sin \theta}{g}$

Step 4 Use that time in the horizontal column. Substitute the expression for t , not a number.

Why #2 again: now you know t and want Δx . #2 is the one with $x - x_0$ and t in it, and with $a_x = 0$ it becomes just $\Delta x = v_{x0} t$.

$$x - x_0 = v_{x0} t + \frac{1}{2} a_x t^2 \rightarrow R = v_{x0} t \text{ (} a_x = 0 \text{)}$$

$$R = (v_0 \cos \theta) \frac{2v_0 \sin \theta}{g} = \frac{v_0^2 (2 \sin \theta \cos \theta)}{g} \text{ and } 2 \sin \theta \cos \theta = \sin 2\theta, \text{ so } R = \frac{v_0^2 \sin 2\theta}{g}$$

Step 5 Now plug in numbers with units.

$$R = \frac{(28.0 \text{ m/s})^2 \sin(2 \times 30.0^\circ)}{(9.80 \text{ m/s}^2)} = \frac{(784 \text{ m}^2/\text{s}^2)(\sin 60.0^\circ)}{(9.80 \text{ m/s}^2)} = 69.3 \text{ m}$$

(b) Hang time (5 points)

Step 1 It is the landing time from Step 3.

$$t = \frac{2v_0 \sin \theta}{g} = \frac{2(28.0 \text{ m/s})(\sin 30.0^\circ)}{(9.80 \text{ m/s}^2)} = 2.86 \text{ s}$$

(c) Maximum height (5 points)

Step 1 At the top the ball stops rising for an instant: $v_y = 0$.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0} t + \frac{1}{2} a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time) ← use this one

Why #3: you know $v_y = 0$ at the top, v_{y0} and a_y , and you want the height. Time is not needed, so use #3, the one without t.

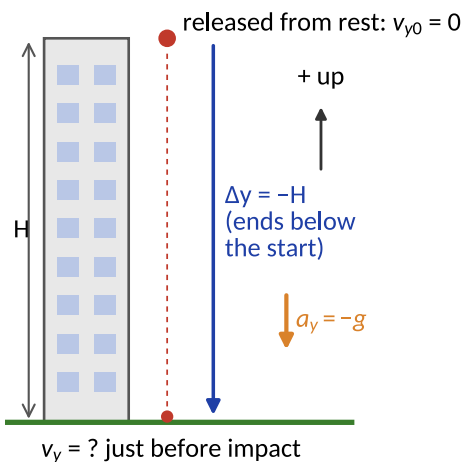
$$v_y^2 = v_{y0}^2 + 2a_y(y - y_0) \rightarrow 0 = (v_0 \sin \theta)^2 - 2g h_{\max}$$

Step 2 Solve with letters, then plug in.

$$h_{\max} = \frac{v_0^2 \sin^2 \theta}{2g} = \frac{(28.0 \text{ m/s})^2 (\sin 30.0^\circ)^2}{2(9.80 \text{ m/s}^2)} = \boxed{10.0 \text{ m}}$$

Check: $v_{x0} = (28.0 \text{ m/s})(\cos 30.0^\circ) = 24.25 \text{ m/s}$, and $R = v_{x0} t = (24.25 \text{ m/s})(2.857 \text{ s}) = 69.28 \text{ m} \approx 69.3 \text{ m}$. ✓

4. Object dropped from a skyscraper (25 points), variables only



No numbers anywhere: every answer is in terms of H and g.

Up is +, origin at the release point. The ground is H below the start, so $\Delta y = -H$.

(a) Signed impact velocity (15 points)

Step 1 List the five variables (vertical, so y in place of x).

v_{y0} (initial)	v_y (final)	a_y	t	$y - y_0 = \Delta y$
0 (from rest)	? (wanted)	-g	not given, not asked	-H

Step 2 Pick the equation.

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy)
#2	$y = y_0 + v_{y0}t + \frac{1}{2}a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time) ← use this one

Why #3: t is not given and not asked, and you want the final velocity. #3 is the only one without t.

Step 3 Solve with letters.

$$v_y^2 = v_{y0}^2 + 2a_y(y - y_0) = 0^2 + 2(-g)(-H) = 2gH \rightarrow v_y = \pm\sqrt{2gH}$$

Step 4 Pick the sign.

The object is moving down and down is negative, so take the negative root: $v_y = -\sqrt{2gH}$

Units check: $\sqrt{(m/s^2)(m)} = \sqrt{m^2/s^2} = m/s$.

(b) Velocity as a function of time, and when it is valid (10 points)**Step 1 Velocity at any time t.**

AP sheet	equation	leaves out
#1	$v_y = v_{y0} + a_y t$	$y - y_0$ (Δy) ← use this one
#2	$y = y_0 + v_{y0}t + \frac{1}{2}a_y t^2$	v_y (final velocity)
#3	$v_y^2 = v_{y0}^2 + 2a_y(y - y_0)$	t (time)

Why #1: you want velocity as a function of time, and Δy is not part of the question. #1 connects v_y and t directly.

$$v_y = v_{y0} + a_y t = 0 + (-g)t \rightarrow \mathbf{v(t) = -gt}$$

Step 2 It only works until the object hits the ground. Find the impact time t_f .

Why #2: at impact you know $\Delta y = -H$, $v_{y0} = 0$ and $a_y = -g$, and you want t. #2 connects those without needing the final velocity.

$$y - y_0 = v_{y0}t + \frac{1}{2}a_y t^2 \rightarrow -H = 0 - \frac{1}{2}g t_f^2 \rightarrow t_f^2 = \frac{2H}{g} \rightarrow t_f = \sqrt{2H/g}$$

Step 3 State the interval.

From release ($t = 0$) to impact: $\mathbf{0 \leq t \leq \sqrt{2H/g}}$

Check: $v(t_f) = -g\sqrt{2H/g} = -\sqrt{2gH}$, the same answer as part (a).

Suggested partial credit (25 points per question, 100 total)

1. Acceleration-time graph

Suggested partial credit	pts
Slope formula $a = \Delta v / \Delta t$ set up with values and units for each of the 4 segments (2 each)	8
Correct slope value with sign and units for each segment (1 each)	4
Four horizontal segments at the correct accelerations (2 each)	8
Changes in acceleration occur at the correct times; no slanted lines	2
Axes labeled with quantity and units; readable scale	3
Total	25

2. Braking car

Suggested partial credit	pts
(a) Chooses sheet #3, $v_x^2 = v_{x0}^2 + 2a_x(x - x_0)$, because time is not involved	3
(a) Correct signs: $v = 0$ and a negative (opposite the motion)	3
(a) Solves symbolically, $\Delta x = -v_{x0}^2 / (2a_x)$, before substituting	4
(a) Substitution with units in every parenthesis	3
(a) Final answer, 3 significant figures, units, boxed	2
(b) Sheet #1 solved for t , substitution with units, answer (2 + 2 + 1)	5
(c) Shows $\Delta x \propto v_{x0}^2$ from the part (a) expression	3
(c) Factor of 4 (not 2)	2
Total	25

3. Football range

Suggested partial credit	pts
(a) Resolves v_0 into $v_0 \cos \theta$ and $v_0 \sin \theta$	2
(a) Uses sheet #2 (y-form) with $\Delta y = 0$ to get $t = 2v_0 \sin \theta / g$	4
(a) $R = (v_0 \cos \theta)t$ with the expression for t substituted (not a number)	4
(a) Simplifies to $R = v_0^2 \sin 2\theta / g$ (or the equivalent $2v_0^2 \sin \theta \cos \theta / g$)	2
(a) Numerical range with units, 3 significant figures, boxed	3
(b) $t = 2v_0 \sin \theta / g$, substitution with units, answer (2 + 2 + 1)	5
(c) Uses sheet #3 (y-form) with $v_y = 0$ at the top	2
(c) $h_{\max} = v_0^2 \sin^2 \theta / (2g)$	1
(c) Substitution with units and answer	2
Total	25

4. Object dropped from a skyscraper

Suggested partial credit	pts
(a) States $v_0 = 0$, $a = -g$ and $\Delta y = -H$ consistent with up positive	4
(a) Uses sheet #3 (y-form), since t is not involved	3
(a) Algebra to $v^2 = 2gH$	3
(a) Chooses the negative root with a reason (moving down)	3
(a) Final expression $v_y = -\sqrt{2gH}$, boxed	2
(b) $v(t) = -gt$ from sheet #1	4
(b) Impact time $t = \sqrt{2H/g}$ from sheet #2 with $\Delta y = -H$	3
(b) Interval $0 \leq t \leq \sqrt{2H/g}$ stated	2
(b) Consistency with part (a) (or other check)	1
Total	25